that might have applied to the first billionth of a billionth of a second of the universe's existence. Others study modifications of Einstein's theory of gravity that could pertain to objects as hypothetical as black holes that can only exist in higher dimensions of space. Some cosmologists even study whole hypothetical universes that are very explicitly not our own—universes in which the cosmos has a totally different shape, number of dimensions, and history—in order to gain insight into the mathematical structure of theories that *might* someday be found to have relevance to us.[fn5]

The upshot of all this is that cosmology means a lot of different things to a lot of different people. A cosmologist who studies the evolution of galaxies might be utterly lost talking with a cosmologist who studies the way quantum field theory can make black holes evaporate, and vice versa.

As for me, I love it all. I first learned cosmology was a thing when I was about ten years old, through encounters with books and lectures by Stephen Hawking. He was talking about black holes and warped spacetime and the Big Bang and all sorts of stuff that made me feel like my brain was doing backflips. I *could not get enough*. When I found out that Hawking described himself as a cosmologist, I knew that was what I wanted to be. Through the years, I've done research across the whole range, bouncing back and forth between physics and astronomy departments, studying black holes, galaxies, intergalactic gas, intricacies of the Big Bang, dark matter, and the possibility that the universe might suddenly blink out of existence.[fn6] I even dabbled in experimental particle physics for a while, in my misspent youth, playing with lasers in a nuclear physics lab (despite what the records might say, the fire was not my fault) and paddling an inflatable boat around a 40-meter-tall water-filled underground neutrino detector (that explosion was not my fault either).

These days, I'm pretty solidly a theorist, which is probably better for everyone. This means I don't carry out observations or experiments or analyze data, though I do frequently make predictions for what future observations or experiments might see. I work mainly in an area physicists call phenomenology—the space between the development of new theories and the part where they're actually tested. That is to say, I find creative new ways to connect the things the fundamental-theory people hypothesize about the structure of the universe with what the observational astronomers and experimental physicists hope to see in their data. It means I have to learn a lot about everything,[fn7] and it's a heck of a lot of fun.